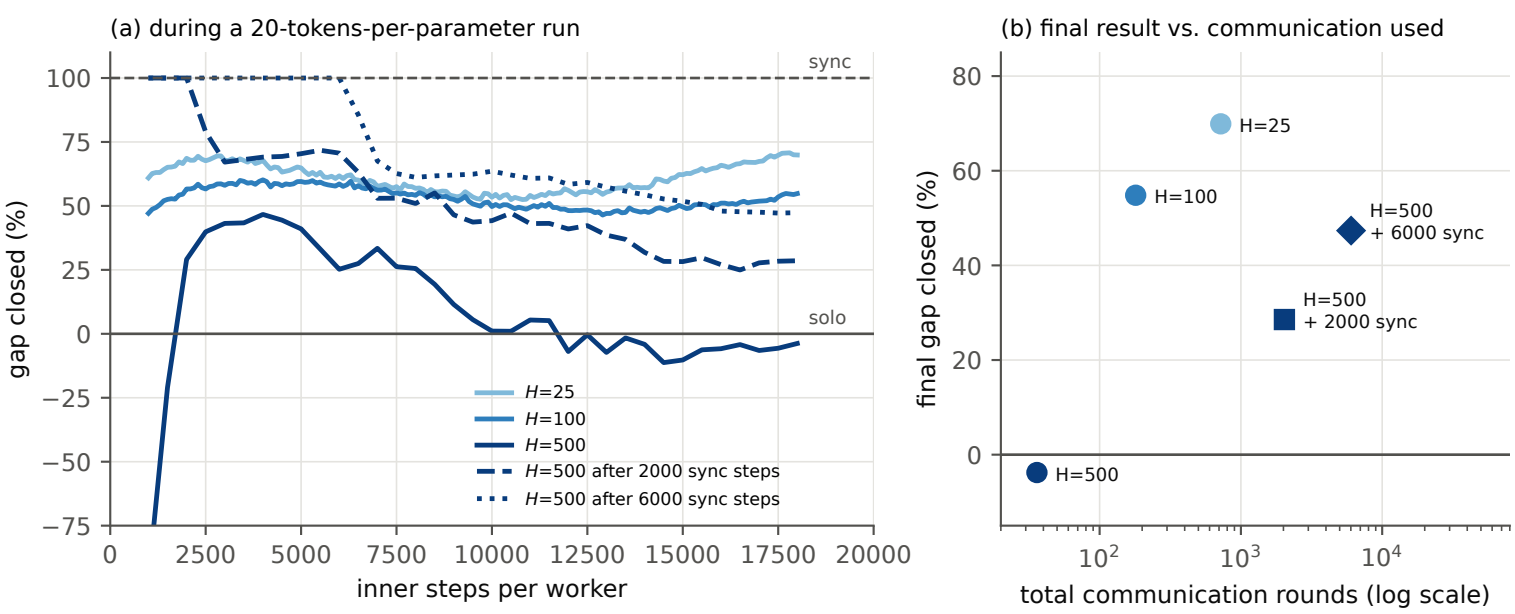
(a) during a 20-tokens-per-parameter run
sync
solo
gap closed (%)
100
75
50
25
0
−25
−50
−75
0
2500
5000
7500
10000
12500
15000
17500
20000
inner steps per worker
H=25
H=100
H=500
H=500 after 2000 sync steps
H=500 after 6000 sync steps
(b) final result vs. communication used
final gap closed (%)
80
60
40
20
0
H=25
H=100
H=500
+ 6000 sync
H=500
+ 2000 sync
H=500
10^2
10^3
10^4
total communication rounds (log scale)